

What Performance Measures Do Real Estate Investors Chase?

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Abstract: We examine how retail and institutional real estate investors benchmark fund performance across a comprehensive set of performance metrics. Our analysis focuses on a subset of real estate funds for which fund flows, investor type (retail vs. institutional), and risk-adjusted performance are observable. We find that benchmarking practices differ systematically by investor type. Retail investors respond primarily to raw, unadjusted returns, whereas institutional investors are most sensitive to risk-adjusted performance, particularly CAPM alphas estimated using a real estate index as the market benchmark. Although multifactor models most accurately explain historical fund performance, they exert limited influence on the investment decisions or benchmarking practices of either investor group. Ultimately, what drives performance may not be what drives capital.

Keywords: Real Estate Investment, Investor Sophistication, Fund Flows, Retail Investors, Institutional Investors, Performance Chasing, Fund Performance, Alpha, Real Estate Mutual Fund (REMF)

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1. Introduction

Commercial real estate is widely recognized as a vital component of a well-diversified investment portfolio. Investors seeking exposure to this asset class face a broad array of options, ranging from direct investments in individual properties and locally managed private vehicles to large, institutionally sponsored funds offered by global asset managers. These vehicles vary in scope, targeting specific properties, locations, or offering broader commercial real estate exposure. Additionally, investors can access the sector through securitized vehicles such as public and private real estate investment trusts (REITs). While the literature on real estate investment is extensive, meaningful gaps remain—particularly concerning investor profiles, sponsor strategies, fund characteristics, and even the basic methodology used to calculate returns.

A well-developed body of financial research addresses the estimation of risk-adjusted returns in public equity markets (Sharpe, 1964; Fama and French, 1993; Carhart, 1997). However, most real estate investments occur through private vehicles, where the lack of transparency and limited liquidity pose significant challenges to risk adjustment (Seah et al., 2025; Sleptcova and Falkenbach, 2020; Devine et al., 2022; Downs et al., 2015). The industry categorizes real estate strategies into broad buckets—core, core-plus, value-add, and opportunistic—under the general premise that higher risk should correspond to higher expected returns (Fisher and Hartzell, 2016). However, private fund structures vary considerably in their fee arrangements (e.g., carried interest, preferred return structures) and in the return metrics used, which range from simple cash-on-cash yields to more complex internal rate of return (IRR) measures. This heterogeneity in performance measurement contributes to a relatively thin academic literature on how commercial real estate investors—particularly across investor types—benchmark fund performance.

Recent academic work has begun to examine fund selection and incentive structures in private real estate vehicles, though the literature remains limited. Jackson, Ling, and Naranjo (2023) find that institutional investors tend to benchmark performance using smoothed returns reported by private real estate funds. Li and Riddiough (2023) document that underperforming private funds exhibit high survival rates, raising the possibility that institutional investors are either constrained in their ability to evaluate performance or are optimizing for objectives beyond return maximization. Yu et al. (2025) focus on investor networks and how they affect fund performance. While these emerging studies offer important insights into a segment of the market that accounts for substantial capital allocation, they are ultimately constrained by data limitations. Reported net asset values (NAVs) and internal rates of return (IRRs) are often imprecise and subject to manipulation, as also noted by Jackson, Ling, and Naranjo (2023).

This study extends the real estate fund literature by focusing on a distinct subset of investment vehicles: public real estate mutual funds (REMFs). Although REMFs represent a relatively small share of total real estate investment, they offer several advantages over private funds for studying investor behavior. First, REMFs are marked to market daily based on publicly traded real estate securities, enabling the calculation of both raw returns and a range of risk-adjusted performance measures commonly used in the finance literature. Second, capital flows are observable monthly, providing a direct measure of marginal investor allocation decisions. Third, REMFs offer distinct retail and institutional share classes, allowing for the analysis of investor behavior by type. Taken together, these features facilitate the empirical investigation of a novel question in the real estate literature: What performance metrics do investors chase?

In recent years, the gap between how mutual fund managers should be evaluated and how they are actually evaluated has motivated researchers to examine investor behavior in fund

selection. A well-established literature documents that mutual fund investors chase past performance (Chevalier & Ellison, 1997; Chou & Hardin, 2014; Ling & Naranjo, 2006; Sialm et al., 2015, among others), yet the specific performance metrics that investors respond to remain relatively underexplored. Evidence from Barber, Huang, and Odean (2016) and Berk and van Binsbergen (2016) suggests that investors in general equity mutual funds respond to CAPM alphas, indicating a tendency to discount market risk in their evaluation of fund managers. Subsequent studies further develop this view, portraying mutual fund investors as sophisticated agents who allocate capital to alpha-generating managers (Li et al., 2017; Back, Crane, & Crotty, 2018; Artavanis et al., 2019).

Concurrently—and in contrast to the view of investors as sophisticated allocators of capital—Ben-David et al. (2022) find that mutual fund investors primarily respond to Morningstar ratings. While these ratings are based on risk-adjusted performance, the authors demonstrate that investors react to the star ratings themselves without exhibiting an understanding of the underlying methodology. Moreover, their simulation evidence casts doubt on the conclusions of Barber, Huang, and Odean (2016) and Berk and van Binsbergen (2016), suggesting that investor sensitivity to CAPM alpha may be spurious. As a result, the debate over the degree of investor sophistication in the mutual fund literature remains unresolved, with ongoing disagreement across empirical findings.

This study focuses on the performance metrics that investors in real estate mutual funds (REMFs) respond to and use for benchmarking. In the broader mutual fund literature, benchmark models such as the Capital Asset Pricing Model (CAPM), the Fama–French Three-Factor model (FF3), and the Carhart Four-Factor model (FFC4) are widely used to estimate alpha (Fama & French, 1993; Carhart, 1997). However, how best to benchmark REMF performance remains less

well defined. REMFs primarily invest in publicly traded Real Estate Investment Trusts (REITs), yet the literature evaluating REMF performance is relatively limited and employs a diverse range of benchmarking approaches. These include applying traditional factor models developed for diversified equity funds, adapting those models using the REIT universe, or augmenting standard models with real estate-specific factors. Notably, the choice of benchmark model can substantially affect conclusions regarding whether REMF managers generate alpha or even track the market effectively.

A recurring theme in the REMF literature is the use of heterogeneous benchmark models, which frequently combine traditional asset pricing factors with REIT-specific factors. Unsurprisingly, the choice of benchmark often drives the conclusions regarding fund and manager performance. For instance, Kallberg, Liu, and Trzcinka (2000), one of the earliest studies on REMFs, found that these funds outperform on average—a result that contrasts with findings for diversified mutual funds. Their custom multifactor model included traditional size and value factors, a bond factor, and a REIT-based market factor. In contrast, other studies report no evidence of outperformance (Chou & Hardin, 2014; Lantushenko & Nelling, 2019; Hartzell et al., 2010). Additional papers adopt hybrid models combining traditional and REIT-based factors, yielding mixed results. For example, Derwall et al. (2009) employed a traditional Fama–French three-factor model but incorporated a momentum factor derived solely from REIT returns. We include in the appendix a list of papers examining REMFs and the benchmark models used. Collectively, the literature lacks consensus on both appropriate benchmarking methods and the benchmarks that investors themselves rely on in evaluating REMFs.¹

¹ There exists a relatively wide usage of factor models to explain REIT returns themselves. Letdin et al. (2020) provide an excellent review and Letdin, Seagraves, and Sirmans (working paper) provide a thorough analysis of REIT-based factors to explain returns.

This study offers a distinct contribution by focusing on the benchmark models that investors *actually* use when allocating capital rather than those they *should* use according to theoretical frameworks. In line with the mainstream finance literature, we examine the relationship between various performance measures and subsequent fund flows. Using a comprehensive sample of real estate mutual funds (REMFs) from 1995 to 2022, we construct a range of performance metrics to assess investor responsiveness. These include simple, commonly available indicators such as unadjusted returns and Morningstar star ratings, as well as more complex measures derived from one-, three-, and four-factor models. We estimate these models using both traditional equity factors and alternative specifications based on REIT returns, allowing us to test which performance metrics most effectively explain investor capital allocation behavior.

Our baseline results indicate that investors respond to a combination of unadjusted returns, Morningstar ratings, and CAPM-based alphas. These results suggest that more complex performance measures—such as alphas derived from Fama–French three-factor (FF3) or Carhart four-factor (FFC4) models—are largely ignored by both retail and institutional investors. While informative, these findings provide limited insight into the heterogeneity of investor behavior across investor types. We construct a proxy variable that classifies each fund as predominantly retail- or institutional-facing based on share class characteristics to address this limitation. This allows us to investigate whether benchmark preferences differ systematically by investor type. We hypothesize that institutional investors are more likely to be risk-aware, and therefore, more responsive to risk-adjusted performance measures than retail investors.

When we partition the sample into retail- and institutional-oriented funds, the results offer clearer insights into investor behavior. For retail-oriented funds, unadjusted returns emerge as the dominant driver of fund flows. In contrast, flows into institutionally oriented funds are primarily

influenced by CAPM alphas computed using a REIT-based market factor. While we do not suggest that all retail investors respond exclusively to raw returns or that all institutional investors rely solely on CAPM-based performance, the findings reveal a distinct preference pattern. Specifically, the results indicate an implicit hierarchy in the performance metrics investors use to evaluate real estate mutual funds.

Our findings have important implications for both academic research and investment management in the real estate space. We document that retail investors in real estate mutual funds (REMFs) tend to rely on the simplest available benchmarks, with unadjusted returns exerting the greatest influence on their capital allocation decisions. In contrast, institutionally oriented funds—which account for approximately 43% of total fund flows—exhibit greater sensitivity to risk-adjusted performance, particularly CAPM alphas based on REIT-specific market indices. These results offer a more nuanced understanding of investor behavior in real estate mutual funds and challenge prior assumptions in the literature that implicitly treat mutual fund investors as uniformly rational and performance-sensitive (e.g., Chou & Hardin, 2014).

The extent to which findings from REMFs generalize to the broader universe of real estate investments is not immediately clear. Nonetheless, the observed lack of sophistication among retail investors aligns with evidence from other segments of the real estate fund landscape. For instance, Riddiough and Wiley (2022) study a sample of unlisted REITs targeting retail investors with modest wealth and find compelling evidence that these investors are prone to suboptimal decisions, often driven by poor financial advice. Conversely, the evidence of greater sophistication among institutional investors in the REMF context presents an interesting contrast to recent findings in the private real estate market. Li and Riddiough (2023) report that institutional investors in private equity real estate frequently fail to identify skilled managers and make suboptimal allocation

decisions. This divergence raises important questions about the contexts in which institutional sophistication manifests and the conditions under which it fails to do so.

In practice, real estate mutual fund (REMF) managers are typically compensated as a function of assets under management (AUM), with increased fund flows driving higher AUM and, in turn, greater management fees. This compensation structure creates strong incentives for fund managers to pursue strategies that maximize inflows, even if such strategies are not aligned with long-term investor interests. Our findings indicate that in retail-oriented REMFs, unadjusted returns are the most salient performance metric influencing investor behavior. As a result, managers of these funds are likely to emphasize maximizing short-term raw returns, as doing so is most effective in attracting capital from retail investors. This dynamic can lead to suboptimal outcomes for investors, as it may incentivize excessive risk-taking without commensurate accountability, particularly in the absence of robust performance benchmarking or risk monitoring by retail investors.

In contrast, flows into institutionally oriented funds are primarily driven by risk-adjusted performance. CAPM alpha calculated using REIT-based benchmarks appears to be the most utilized model. This suggests a more disciplined incentive environment in which fund managers are rewarded not merely for generating high returns, but for doing so relative to appropriately risk-adjusted benchmarks. Such a structure is more consistent with efficient contracting theories in delegated portfolio management, where sophisticated investors (principals) design compensation and monitoring mechanisms to align manager (agent) incentives with the performance net of risk. The contrast between retail and institutional segments highlights meaningful differences in the quality of governance and performance evaluation across investor types. It underscores the need for stronger mechanisms to protect less sophisticated investors from misaligned incentives.

The remainder of this paper is organized as follows. Section two describes the data. Section three details the empirical methods and results, Section four presents robustness, Section five concludes.

2. Sample and Summary Statistics

Our sample consists of REMF fund-months from January 1995 to March 2022. The monthly return data comes from the CRSP (Center for Research in Security Prices) Survivor-Bias-Free US Mutual Fund Database. Our sample begins in 1995 to be consistent with other recent studies on REMFs, including Lantushenko and Nelling (2019), Hartzell et al. (2010), Chou and Hardin (2014), and Cici et al. (2011).

We are careful to ensure our sample consists solely of real estate mutual funds through the following screening method. We keep funds with Lipper objective code “RE”, Lipper class code “RE”, Strategic Insight code “RLE” or CRSP objective code “ESDR”. Funds where none of the Lipper or Strategic Insight codes were available are excluded from the analysis. Our focus is on domestic REMFs, so we excluded funds that contained “international,” “emerging market,” “global,” “euro,” “currency,” or “world” in their fund name. We are focused on open-ended actively managed funds. Therefore, we also screened out funds flagged as index funds or ETFs using the “index_fund_flag” and “et_flag” variables in CRSP. We exclude funds with missing or zero expenses ratios. Additionally, we exclude target date funds by eliminating funds that contain “target”, and years from 2005 to 2070 in the name. Finally, we reviewed the final list of funds manually to screen out any leftover funds that are not REMFs and we require at least 2 years of data to conduct our tests. This procedure yielded a total of 115 unique funds and 18,010 fund-month observations.

We then merged this set of funds with Morningstar Direct to obtain Morningstar ratings. We identified Morningstar ratings for 81 of the 115 funds and 10,333 fund-month observations. Given that a meaningful number of funds would be dropped if we excluded funds without Morningstar ratings, we conduct all tests in this paper on the sample with Morningstar ratings (Morningstar sample) and the sample of all funds (full sample).

Our variables of interest for this study are each performance metric, fund flows, and whether the fund is retail or institution-oriented. We define fund flows following the standard convention in the literature:

$$Flow_{i,t} = \frac{TNA_{i,t}}{TNA_{i,t-1}} - (1 + R_{i,t}) \quad (1)$$

Where TNA is the total net assets, i is the fund and t is the month, and R is the fund return in month t . Morningstar ratings are the ratings, 1 to 5, in a given fund-month. For each return and alpha variable, we calculate them on both the net and gross basis. Gross returns are net returns plus the expense ratio divided by 12. For each benchmark model, we estimate alpha using 12-month rolling betas.

We refer to models using factors based on all stocks from Kenneth French's website as *traditional* factor models.² We also compute what we refer to as *REIT-based* factor models. These models use the same definition and breakpoints as Kenneth French's website, but they are created manually within the REIT universe. Specifically, the REIT-based market factor is constructed using the NAREIT index. The size portfolio is constructed by splitting all REITs into two groups based on median market capitalization in June of each year. The size return is equal to the small

² https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. We Thank Ken French for making this data available.

portfolio return minus the large portfolio return. The book-to-market portfolio is formed by dividing REITs into three groups based on their book-to-market ratios as of December each year. The low book-to-market portfolio is the bottom 30%, and the high book-to-market is the top 30%. The difference between the high and low book-to-market portfolios is the portfolio return. Lastly, the momentum factor is constructed by sorting all REITs based on their month t-2 to t-12 cumulative returns. Winners (top 30% momentum REITs) minus losers (bottom 30% momentum REITs) is how the factor is created. For traditional factor models, we refer to the CAPM, Fama French 3-factor model, and Fama French Carhart 4-factor model as CAPM, FF3, and FFC4, respectively. For each REIT-based version, we add “REIT” to the same abbreviations.

Next, we construct a measure to capture whether a fund is retail- or institution-oriented. Following the mutual fund literature, if a fund has multiple classes (e.g., A, B, and C), we aggregate them for our study. However, CRSP provides an identifier at the class level that identifies whether the fund is a retail fund or an institutional fund. We use these indicators weighted by total net assets (TNA) at the class level to calculate a percentage of retail orientation at the fund level. To illustrate, if a fund has a Class A – Retail Class with \$100 million of TNA and a Class B – Institutional Class with \$20m of TNA, our retail-oriented measure equals 83.33% ($100/(100+20)$). We define any fund greater than 50% as retail-oriented and other funds as institutional-oriented. Using this process, we have 10,333 retail-oriented funds and 7,677 institution-oriented funds. For the remainder of the paper, we will refer to retail-oriented funds as *retail funds* and institution-oriented funds as *institutional funds*. We have six different sub-samples used throughout the study: First, we have a sample of funds that have Morningstar ratings and a full sample that includes all funds. Then, within the Morningstar and full sample, we examine retail funds, institutional funds, and both combined.

Table 1 presents summary statistics for each variable used throughout the study. Panel A displays statistics for the combined sample of both retail and institutional funds. The typical Morningstar rating is 2.88, and the median is 3.0. One statistic of note is that the average fund flows are negative over the sample period. This reflects the overall decline in active REMF popularity, particularly in recent years. Given that REITs continue to grow at a steady pace, it potentially reflects an overall trend toward passively managed funds rather than active ones. As expected, net returns are lower than gross returns by approximately ten bps per month. Panel B and C present summary statistics for the retail and institutional funds, respectively. There are moderate differences between each group. Institutional funds have more significant overall outflows during the sample period compared to retail ones. Additionally, net alphas for the REIT-based factor models for retail funds are negative on average. In contrast, each of them is positive for the institutional funds. This is very limited evidence that could suggest institutionally oriented funds could modestly outperform on a risk-adjusted basis. For the remainder of the analysis, we present results using net returns and alpha performance metrics for brevity. The analysis using gross returns and alpha performance metrics are qualitatively similar.

[Insert Table 1 Here]

Table 2 displays the correlations between the various performance measures. Returns and alphas are the average of the past 12 months. Correlations are of concern because if the performance metrics are sufficiently highly correlated, differentiating between metrics is not a fruitful question. Interestingly, given many models use overlapping factors, we observe only moderate correlations across metrics. For example, the correlation between returns and Morningstar ratings is merely 0.07. The only correlation greater than 0.5 for net returns is the traditional CAPM at 0.61.

[Insert Table 2 Here]

It is equally interesting to observe the correlations between traditional factor models and their REIT-based counterparts. Again, if they are highly correlated, the effort to calculate factor models within REITs is futile. However, this is far from the case. The correlation between traditional CAPM and CAPM REIT is 0.33. The correlations between FF3 and FF3 REIT and FFC4 and FFC4 REIT are 0.20 and 0.08, respectively. These relatively low correlations suggest a significant difference between factor models using all stocks and factor models constructed within the REIT universe. Now that we have established that performance metrics vary, we next investigate REMF performance and which performance metrics investors chase.

3. Empirical Methods and Results

3.1 Univariate Analysis

We begin the empirical analysis by observing the average net alphas using each benchmark model for REMFs over the sample period. This serves as an overview of REMF performance and sensitivity to benchmark choice. For each of the alphas, we annualize them by multiplying the monthly average by 12. Figure 1 presents the results, with Panel A displaying the combined sample and panels B and C displaying the retail and institutional funds, respectively. Whether REMFs beat the market on average depends heavily on the benchmark model.

[Insert Figure 1 Here]

There is significant variation in average alphas across benchmark models. The most popular risk-adjusted model by far is the CAPM. This is evidenced by the widespread reporting of CAPM betas across investment websites, academic textbooks, and survey data of corporations (Graham, 2022). Arguably, the next most widespread model is the Fama French 3-factor model,

as evidenced by the common use of size and value vs. growth categories to describe funds by Morningstar and the funds themselves.³ Using these two models with traditional factors, REMFs display impressive average performance. REMFs generate an average of 0.72% and 2.04% in net alpha per year using the traditional CAPM and FF3, respectively. After accounting for the traditional momentum factor, REMF outperformance turns slightly negative. On a net alpha basis, the average FFC4 alpha is -0.24% per year. This result is consistent with the literature that documents that both REMFs and REITs are impacted heavily by momentum (Derwall et al., 2009; Chui et al., 2003a; Letdin et al., 2025).

As Roll (1978) notes, the choice of benchmark is incredibly important. If selected incorrectly, a passive index with no skill could be attributed alpha.⁴ This is particularly notable for REMFs because most of their holdings are REITs. Therefore, the portfolio is quite different from the typical proxy for the market portfolio. A regression of a REIT index on the market factor defined by Fama and French (1993) would result in a positive or negative alpha even though the index was passive.

A cursory view of Figure 1 demonstrates the impact of using REITs to construct CAPM, FF3, and FFC4 factors on average alphas.⁵ Each of the REIT-based alphas are either negative or small on average. The average net alphas using REIT-based factors is -0.12%, 0.24 %, and -0.24%

³ For example, in grouping funds, Fidelity classifies them as “Large Growth,” “Large Value,” “Small-Mid Growth,” and “Small/Mid Value.” Source : <https://www.fidelity.com/mutual-funds/fidelity-funds/overview>

⁴ In the REMF space, Hartzell, Mulhlofer, and Titman (2010) demonstrate that this is the case in the context of whether to use simply a REIT market factor or to use value and size factors constructed by REITs as well. They found that many REMFs tilted their portfolios towards small cap and value REITs, leading to alpha using the CAPM with REIT factors as the benchmark.

⁵ The theoretical support for using REIT-based factor models is sparse. It is largely a function of empirical success. The focus of our study is whether investors use them to make investment decisions.

for CAPM, FF3, and FFC4, respectively. These findings are critically important for understanding whether active managers add value in the REMF space. Crossing the threshold from positive to negative average net alphas is an important cutoff because negative values imply the investor is better off holding the passive index than paying an active manager. Like traditional factor models, adding the momentum factor (FFC4) erodes an economically sizable amount of alpha.

Panels B and C examine net alphas broken out by retail and institutional funds, respectively. As mentioned in the discussion of Table 1, we do observe a notable difference in performance based on REIT-based factor models. While the values are modest, they are directionally opposite. For the retail-oriented funds, the average net alpha for each of the REIT-based factor models is negative. In contrast, the average net alpha for each REIT-based factor model for institutionally oriented funds is positive.

Manager skill itself is more appropriately observed at the gross alpha level. If gross alphas are positive and net alphas are zero, it simply suggests that managers capture their outperformance with fees. Regardless of the benchmark model, we observe in Table 1 that gross alphas are all positive on average. This is consistent with the findings of Chou and Hardin (2014) and Chacon, Morillon, and Kothari (2022). This result suggests that REMFs outperform on a risk-adjusted basis, regardless of the benchmark model. However, when using REIT-based benchmarks, that outperformance is not generally passed through to investors after fees.

3.2 Flow-Performance Analysis

We next examine how investors benchmark performance based on where their capital flows. Following the finance literature, how funds flow following various measures of performance can indicate what investors value (Ben-David et al., 2022; Berk & van Binsbergen,

2016; Barber et al., 2016, among others). The underlying assumption is that funds evaluated positively will attract more capital than funds evaluated negatively.⁶ Various empirical designs have been proposed to evaluate the flow-performance relation for mutual funds using various benchmarks. However, recent findings from Ben-David et al. (2022) cast doubt on several of these methods as they show that simulated data designed to chase simple returns lead to spurious findings. In light of these findings, we rely on the work of Ben-David et al. (2022) for our empirical tests.

We begin by ranking funds monthly by each performance metric. The performance metrics examined include the traditional and REIT-based factor models alongside unadjusted returns and Morningstar ratings. It should be noted that because these are ranked within a month, ranking by unadjusted returns and ranking by benchmark-adjusted returns would lead to an identical set of fund rankings.⁷ Given returns and alphas are volatile month to month and investors tend to note performance over a longer period, we take the average return and average alpha over the past 12 months.⁸ Funds on standard websites such as Fidelity or Blackrock typically present historical performance in 1-year, 3-year, 5-year, and 10-year horizons. We take the ranked funds by each performance metric and sort them into quartiles. Top-rated funds are those in the top quartile of performance, and bottom-rated funds are those in the bottom quartile.

⁶ Ling and Naranjo (2003; 2006) focus on the dynamic relation between flows and returns in REITs and REMFs. Rather than study the wide array of variables that may impact fund flows and returns, our study is narrowly focused on the differential fund flow outcomes that follow various performance measures. We assume these underlying dynamics would be present for all performance measures and, therefore, not impact the differences we observe between them.

⁷ If funds had different listed benchmarks, it would be fruitful to examine benchmark adjusted separately. However, almost all of the funds have a standard REIT index as their benchmark. Therefore, ranking by unadjusted returns within a month would produce identical rankings.

⁸ We conduct robustness around this assumption later in the analysis.

Next, we observe average fund flows to the top and bottom-ranked funds and the difference between the two. Table 3 presents the results for both the Morningstar sample and the full sample. Similar to previous tests, Panel A presents retail and institutional funds combined, Panel B examines retail funds, and Panel C examines institution funds. For each sample, we sort the results by the difference in fund flows to top and bottom quartile funds. For Panels A, B, and C, we examine both the full sample of funds and the subset of funds with Morningstar ratings.

For the combined sample in Panel A, we observe that unadjusted returns have the highest difference in fund flows between top and bottom performing funds in the Morningstar sample. For the full sample, CAPM REIT has 0.1% greater difference than unadjusted returns and the traditional CAPM. As is the theme throughout the paper, more complex factor models appear to have the least importance to both retail and institutional investors.

Turning to retail and institutional funds (Panels B and C), we observe that unadjusted returns are the dominant driver of fund flows for retail-oriented funds. CAPM-REIT alphas are the dominant driver of fund flows for institution-oriented funds. The difference in fund flows for top-performing and bottom-performing funds is economically meaningful. In a given month, the difference in flows from the top and bottom-ranked funds by unadjusted returns for retail funds (Panel B, Full sample) is 1.71%. For the institutional funds, the difference in fund flows is 2.14% for funds sorted on CAPM REIT alphas. Thus far, our results suggest that retail investors generally chase unadjusted returns, and institutional investors chase CAPM-REIT alphas. We find little evidence that more complex factor models are utilized significantly by either investor group.

[Insert Table 3 Here]

Given the inherent difficulty in comparing a large number of factor models, further analysis is needed to increase confidence in the results. In our next set of tests, we compare various performance measures more directly to one another. We follow Ben-David et al. (2022) to conduct a series of horse race tests. Taking two performance metrics at a time, we estimate panel regressions of fund flows on indicators of whether a fund is ranked as a top-performing fund by each metric. We control for calendar month fixed effects and cluster standard errors by fund and month. Our focus is on the difference between the two estimated betas. Therefore, we report the difference in the beta coefficients and a t-statistics of whether that difference is significantly different from zero. The results are displayed in Table 4. Formatted like the previous result, we have the combined sample in Panel A, retail only in Panel B, and institutional only in Panel C. For regressions that include Morningstar, we use the Morningstar sample. For those without Morningstar, we use the full sample of funds.

[Insert Table 4 Here]

We sort the results based on relations of interest, and several are presented more than once for ease of interpretation. Specifically, we present the regression results of unadjusted net returns, Morningstar, CAPM, and CAPM REIT, each compared to all other models. T-Statistics that are bold indicate statistical significance beyond the 10% threshold.

The benchmark models with the most statistical significance relative to other models are the unadjusted (net) returns and CAPM REIT. Fund flows to unadjusted returns are significantly more sensitive to unadjusted returns than the FF3 and FFC4 models using both regular factors and REIT-based factor construction. Similarly, fund flows to CAPM REIT alphas are statistically more sensitive than those to the FF3 and FFC4 models. We do not observe statistical significance when

comparing unadjusted returns to Morningstar and the CAPM models. “Morningstar vs. All” and “CAPM vs. All” do not have any statistically different betas.

Moving to retail funds in Panel B, we observe a more consistent trend. The difference in betas between net returns and every other model is positive, and all but Morningstar are statistically significant. Notably, the difference between net returns and CAPM REIT (the most dominant measure in the institutional sample) has a t-statistic of 2.01. Similar to Panel A, comparing any of the simpler models to more complex factor models consistently demonstrates simpler models are used. Interestingly, the benchmarks that arguably best describe REMF returns (REIT-based FF3 and FFC) are the least utilized by investors.

Finally, we examine the institutional sample in Panel C. In contrast to the retail sample and consistent with our previous results, CAPM REIT is consistently the most dominant model. Critically, comparing CAPM REIT to net returns, we observe the difference in betas is statistically significant. Table 4 further confirms our hypothesis that retail investors tend to chase simple returns, and institutional investors tend to chase CAPM REIT.

3.3 Aggregate Fund Flow Analysis

Thus far, each of our tests has centered around how top-performing active REMFs attract capital relative to other active REMFs. However, the total capital allocated to the real estate active fund industry also varies. In the next set of analyses, we examine how the average performance of active REMFs attracts capital over time. We begin by aggregating our fund-month observations into month observations by equal-weight averaging overall performance using each metric. We then sort monthly performance into quartiles and examine fund flows using two measures: *positive flow* is the percentage of funds in a given month with positive fund flows, and *fund flow percent*

is simply the aggregated percentage of fund flows. If a performance metric is being followed by investors, we would expect to see more fund inflows to the active REMF industry when performance by that metric is strong. The results of these tests are presented in Figure 2.

[Insert Figure 2 Here]

For brevity, we present the results of the retail (Panel A) and institutional (Panel B) samples only and compare only the Unadjusted Return and REIT CAPM. First, unadjusted returns have a consistent and positive upward trend for *fund flow percent* across quartiles. This suggests that as REMFs have higher returns, retail funds attract greater capital flows. Interestingly, this relation for CAPM REIT performance of retail funds is flat or negative. There does not appear to be a relation between CAPM REIT-based performance at the industry level and capital flows to retail funds.

Panel B presents the results for the institutional sample. Unlike the retail sample, the unadjusted returns charge has no discernable pattern. The greatest flows occur in the first (lowest) quartile of performance, and the lowest flows occur in the second quartile. Turning to the REIT CAPM, there is a clear upward trend. Both *fund flow percent* and *positive flow* are monotonically increasing across quartiles. These figures provide further evidence that retail investors tend to chase returns, and institutional investors are primarily concerned with REIT CAPM alphas.

To conclude our analysis of aggregate fund flows, we run time series regressions where the dependent variable is monthly aggregate percentage flows to active REMFs. The independent variables are the various performance metrics. To facilitate comparison, we standardize the independent variables to have a mean of 0 and a standard deviation of 1. Like the horse races in Table 4, we compare the two simultaneously. The specifications include calendar-month fixed effects, and the T-statistics are based on Newey-West standard errors with six lags. For brevity,

we examine the retail (Table 5, Panel A) and institutional samples (Table 5, Panel B) separately and focus on unadjusted returns compared to the rest for retail and REIT CAPM compared to the rest for institutional.

[Insert Table 5 Here]

For the retail sample, net returns have a positive and statistically significant relation to fund flows when included alone (column 1) and included with every other performance metric except the CAPM, where neither load significantly. Importantly, when returns and REIT CAPM are included together in the retail sample, net returns continue to load. In the institutional sample, CAPM REIT loads significantly in isolation and when combined with any other metric. Net returns are indistinguishable from zero (column 3). These time series results, taken together, provide further confirmatory evidence that retail investors chase returns and institutional investors chase CAPM REIT alphas.

4. Robustness

Inherent in the research design are several choices made by the researchers that could impact the results presented thus far. In this section, we aim to mitigate these concerns by conducting a battery of robustness tests. First, we address the cutoff used to identify institutional and retail-oriented funds. For the results thus far, we identify a fund as retail (institutional) oriented if over 50% of the TNA is in a retail (institutional) class. While this seems a logical selection, our results may change significantly with a change in the cutoff point.

Table 6 presents the results of the horse race regressions found in Table 4 using cutoffs of 40%, 50%, and 60%. For brevity, we report the Net Returns relative to other performance metrics for the retail sample (Panel A) and CAPM REIT relative to other performance metrics for the

institutional sample (Panel B). Overall, the results look quite like our baseline findings using the 50% cutoff. In Panel A, we observe that Net Returns continue to be the metric with the statistically dominant impact on fund flows relative to almost all the other metrics. Notably, compared to CAPM REIT, Net Returns continue to have T-statistics consistently in the 2.0 range.

[Insert Table 6 Here]

Moving to Panel B, we observe similar robustness comparing CAPM REIT to other metrics for institutional funds. The difference in the coefficients for the horse race is positive in every case, and statistical significance is present for most comparisons. Comparing CAPM REIT to Net Returns, we observe a positive and statistically significant difference for each cutoff selection.

Next, we move to the time series results using the various cutoff points. The results are presented in Table 7. The comparable baseline result is in Table 5. For brevity, we report Net Returns relative to Morningstar and CAPM REIT for the retail sample (Panel A), and we report CAPM REIT relative to Morningstar and Net Returns for the institutional sample (Panel B). For the retail sample, we observe a positive and significant coefficient of Net Returns for each specification. For the institutional sample, the coefficient on CAPM REIT is positive and statistically significant for all cutoff points. There is no statistical significance in the coefficients for Net Returns or Morningstar in the institutional sample. Taking these results together, we provide robust evidence that the 50% cutoff selection does not meaningfully impact the conclusions drawn. Both the horse race and time series tests are qualitatively similar using 40%, 50%, and 60% cutoffs to select retail and institutionally oriented funds.

[Insert Table 7 Here]

Next, we address the method used to calculate the various alphas. Two primary choices must be made to calculate the alphas examined in the study. First, alphas are calculated using a rolling window ranging from 12 to 36 months of data. There are several tradeoffs to consider when selecting the rolling window. Longer windows have more data and, therefore, more statistical precision. However, longer windows require using potentially stale data and constrain the number of observations in the study. Our baseline results use 12-month rolling windows to maximize the number of observations and focus on the most recent data.

The second design choice required to calculate alphas is over what number of months to average alphas. This choice captures the period in which investors evaluate funds. All funds report returns over the past 12 months, and investors naturally think of returns on an annual basis. Therefore, our baseline results average alphas over the past 12 months.⁹ However, there is also reason to suspect investors may be concerned with shorter- or longer-term performance windows.

Our final analysis re-examines the horse race regressions while varying the alpha calculation on the two dimensions. The results are presented in Table 8. For brevity, the retail sample (Panel A) focuses on Net Returns compared to all other metrics, and the institutional sample (Panel B) focuses on CAPM REIT alphas compared to all other metrics. The *Alpha Window* is the rolling window of historical monthly data used to calculate alpha. The table presents results using 12-, 24-, and 36-month rolling windows. The *Average Months* refers to the period alphas are averaged over. We examine 6, 12, and 18 months average alpha. The baseline results reported in our study have an *Alpha Window* value of 12 and an *Average Months* value of 12.

[Insert Table 8 Here]

⁹ Consider, for example, that interest rates are widely quoted on an annual percentage rate (APR) basis.

Compared to our baseline results, we do not observe large differences in significance when varying the *Alpha Window*. Holding *Average Months* constant at 12, we can see that the various *Alpha Windows* continue to be directionally and mostly statistically similar to the base case. This applies to both Panel A (the retail sample comparing Net Returns to all) and Panel B (the institutional sample comparing CAPM REIT to all).

However, we do observe significant differences in results when we vary the period over which alphas are averaged. For both Panel A (retail) and Panel (institutional), The number of significant T-statistics we observe in the 1-month and 6-month periods dwarf those we observe in the 12 and 18-month windows. This result is not particularly surprising given it is natural to assume investors critique funds mostly on their most recent annual performance. Still, there is little evidence that investors discriminate between funds using net returns (for retail) or CAPM REIT alphas (for institutional) over very short windows. Consistent with expectations, the 12-month results are also stronger than the 18-month results.

5. Conclusion

Are real estate investors rational, sophisticated actors that carefully benchmark managers to identify skill? We find evidence that institutional investors typically risk-adjust and retail investors typically do not. Our findings indicate that investors' marginal dollar allocated to retail-oriented REMFs chases funds that have the highest unadjusted returns in the last 12 months. The marginal dollar allocated to institutionally oriented funds chases alpha using a REIT-based CAPM model.

The evidence of institutional sophistication in public real estate funds contrasts with findings from the private real estate literature, where institutional investors are often portrayed as

failing to select skilled managers (Li & Riddiough, 2023). Reconciling this divergence presents a promising direction for future research, particularly in identifying whether institutional investors differ systematically across public and private market settings.

Investor sophistication has long been a central concern in mutual fund research, particularly since most funds fail to generate net-of-fee outperformance. Theoretical models such as those of Berk and Green (2004) and Pastor and Stambaugh (2012) rely on rational capital allocation to explain the persistence of underperformance. Our findings lend partial support to this framework: institutional flows appear consistent with rational learning and performance evaluation, while retail flows do not. The continued success of retail funds, despite a lack of skill-based capital allocation, points to the importance of investor education and disclosure.

Improving financial literacy among retail investors may enhance capital allocation efficiency and reduce the likelihood of performance-chasing behavior based on misleading or incomplete metrics. Further research into the behavioral drivers of retail investment in real estate funds—such as attention, advice channels, and framing—could provide valuable insights into the persistence of these patterns.

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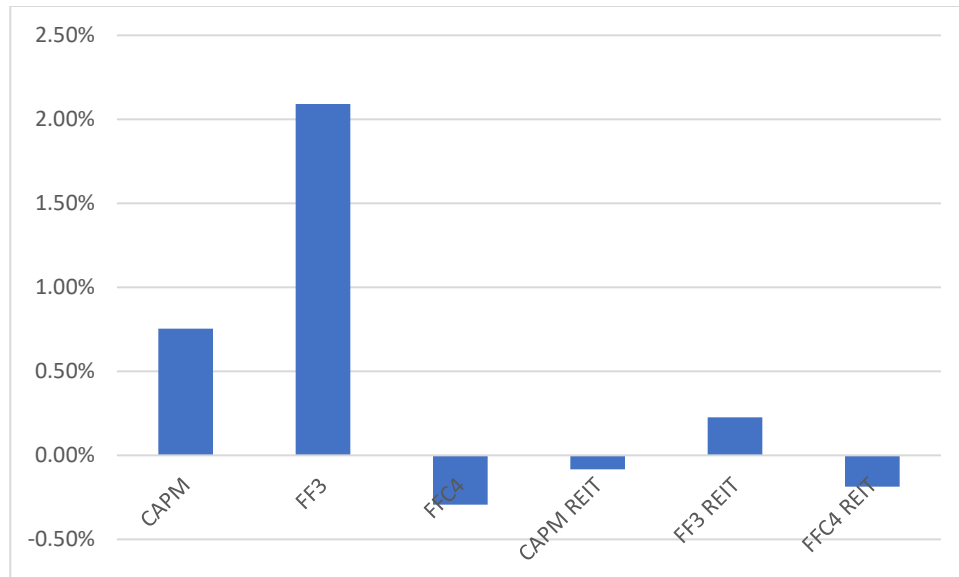
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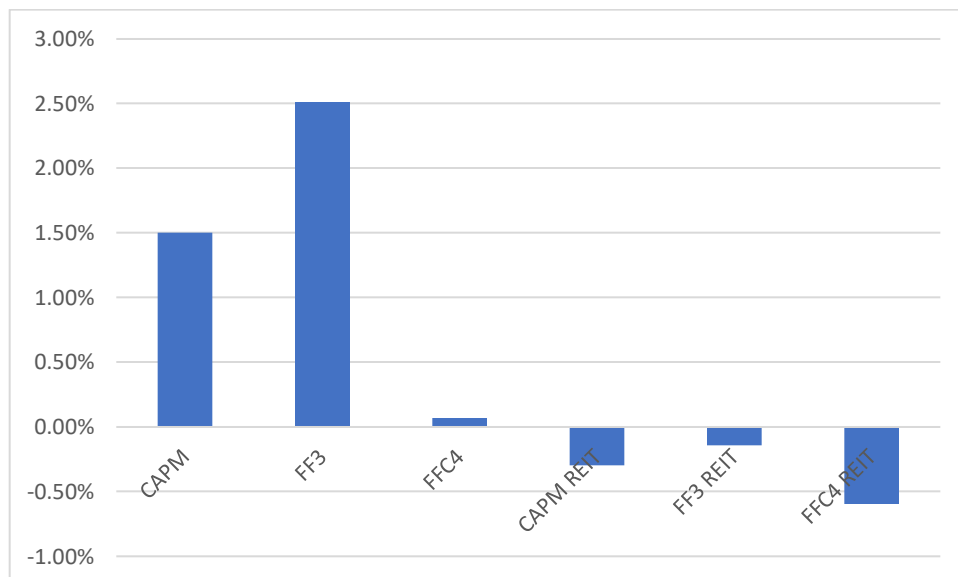
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Figure 1 - Performance by benchmark model

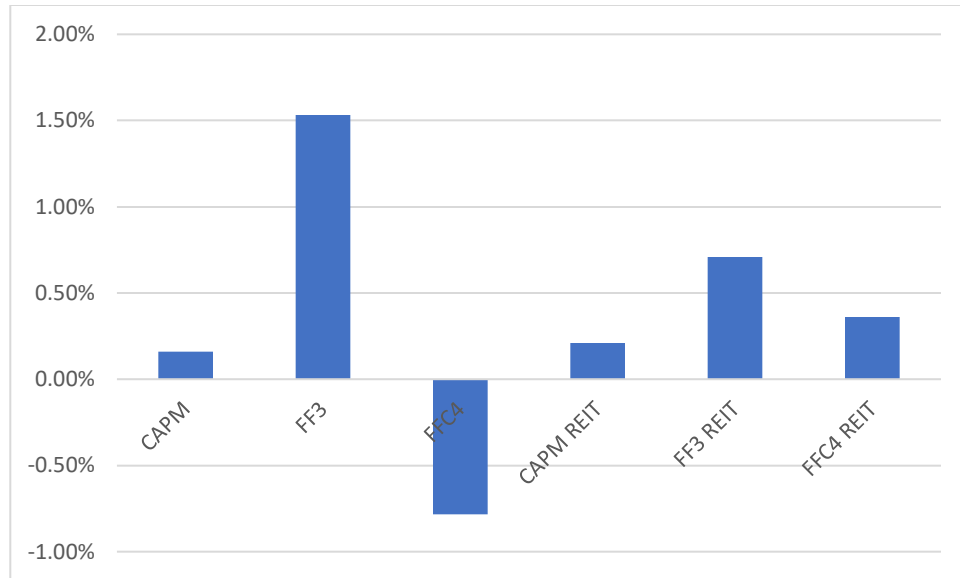
This figure takes the average annualized net alpha in percentage for each factor model. CAPM, FF3, and FFC4 are traditional factor models, and CAPM REIT, FF3 REIT, and FFC4 REIT are the same factor models constructed using REITs rather than all stocks.



Panel A: Combined Sample



Panel B: Retail

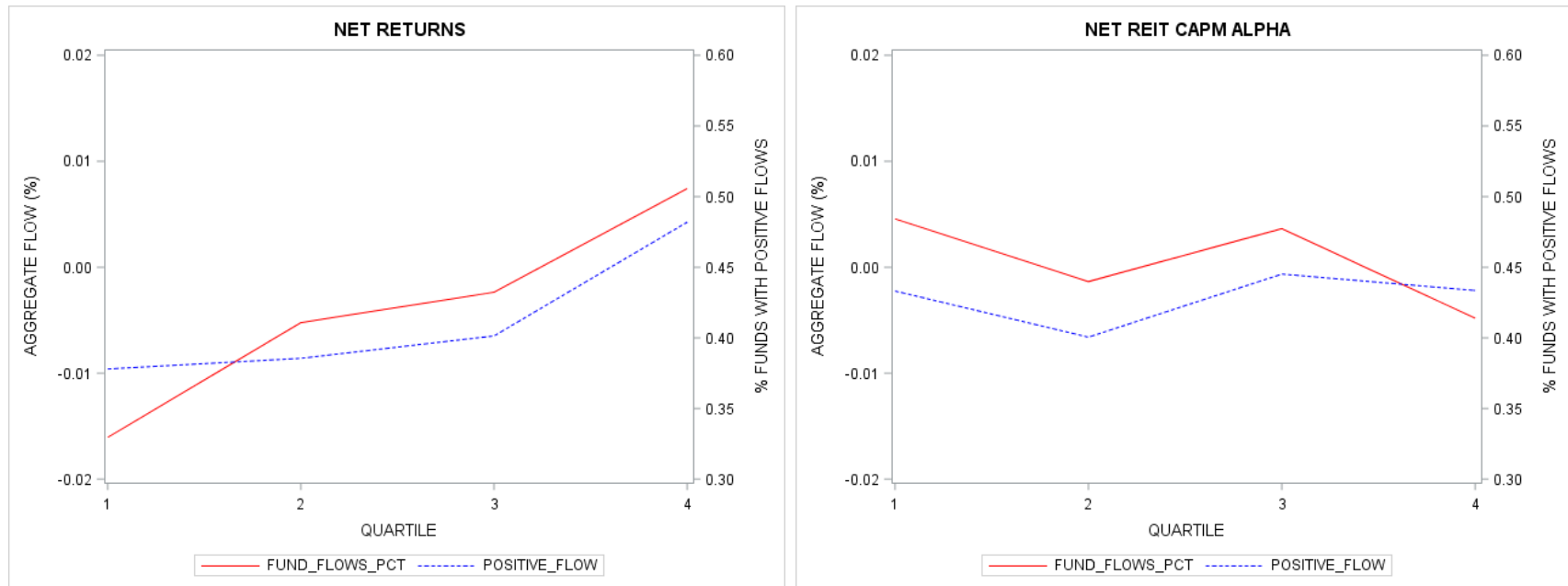


Panel C: Institutional

Figure 2 –Aggregated fund flows

This test examines fund flows at the aggregate time-series level. We take the average of all alphas for each performance measure for a given month. We sort these alphas into quartiles and plot them against average overall fund flows.

Panel A: Retail



Panel B: Institutional

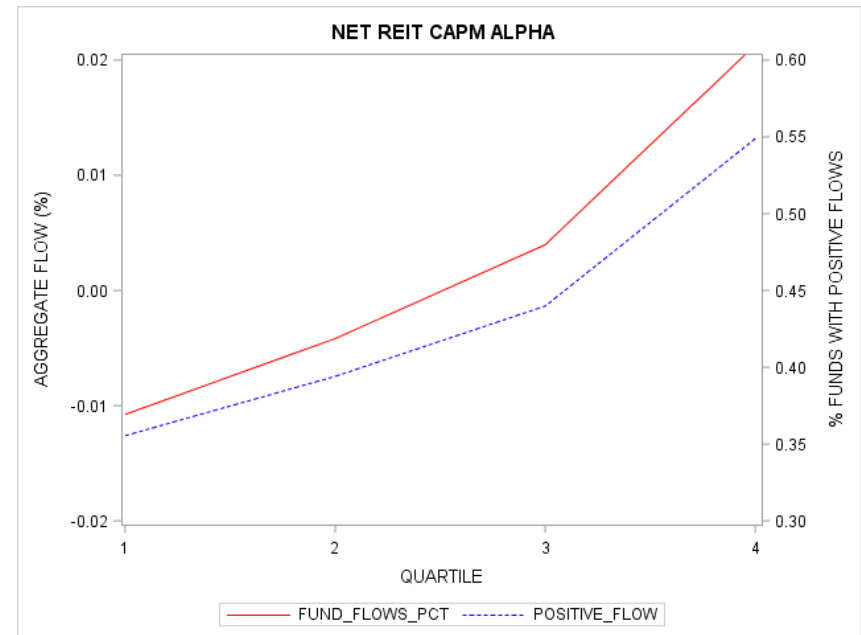
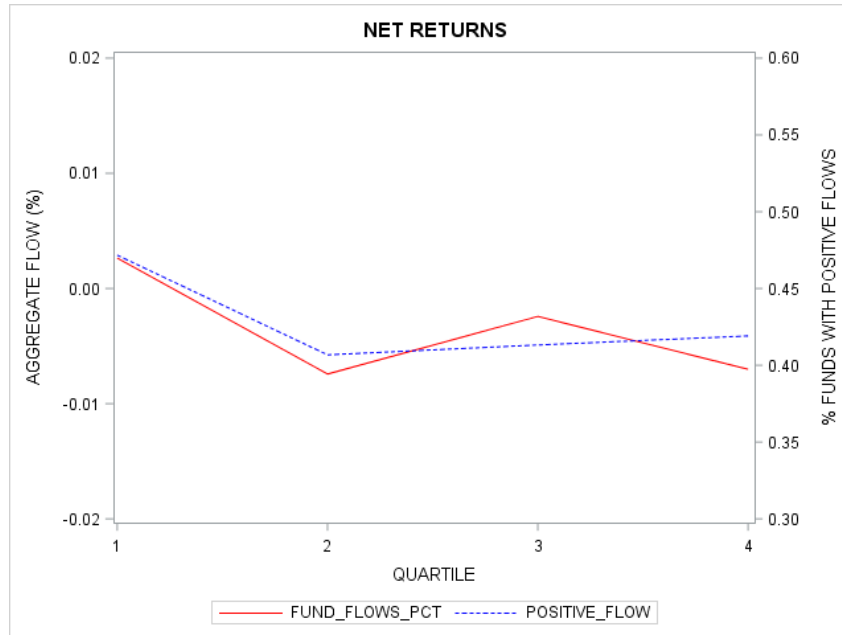


Table 1: Sample statistics

This table presents summary statistics for all variables used throughout the study. CAPM, FF3, and FFC4 refer to traditional factor models as defined in the text, and CAPM REIT, FF3 REIT, and FFC4 REIT refer to REIT-based factor models. Panel A reports the summary statistics for the Full Sample, and Panel B and C report the summary statistics for the Retail Sample and the Institutional Sample, respectively.

Panel A: Full Sample						
Variable	N	Mean	Std Dev	Q1	Median	Q3
Gross Return Variables						
Gross Returns	18,010	0.85%	1.60%	0.16%	0.95%	1.86%
CAPM	18,010	0.16%	1.18%	-0.57%	0.26%	0.98%
FF3	18,010	0.27%	1.13%	-0.46%	0.26%	1.00%
FFC4	18,010	0.07%	1.56%	-1.03%	-0.15%	1.02%
CAPM REIT	18,010	0.09%	0.48%	-0.11%	0.09%	0.29%
FF3 REIT	18,010	0.12%	0.49%	-0.09%	0.10%	0.31%
FFC4 REIT	18,010	0.08%	0.70%	-0.27%	0.10%	0.45%
Net Return Variables						
Net Returns	18,010	0.75%	1.60%	0.06%	0.86%	1.76%
CAPM	18,010	0.06%	1.18%	-0.67%	0.17%	0.89%
FF3	18,010	0.17%	1.13%	-0.56%	0.17%	0.90%
FFC4	18,010	-0.02%	1.56%	-1.12%	-0.24%	0.93%
CAPM REIT	18,010	-0.01%	0.48%	-0.21%	0.00%	0.19%
FF3 REIT	18,010	0.02%	0.49%	-0.19%	0.01%	0.21%
FFC4 REIT	18,010	-0.02%	0.70%	-0.37%	0.00%	0.35%
Fund Flows (%)	18,010	-0.90%	6.44%	-4.36%	-1.32%	2.24%
Morningstar Ratings	10,414	2.88	0.99	2.00	3.00	3.67

Table 1 – Cont'd

Panel B: Retail sample						
Variable	N	Mean	Std Dev	Q1	Median	Q3
Gross Return Variables						
Gross Returns	10,333	0.83%	1.65%	0.14%	0.97%	1.86%
CAPM	10,333	0.23%	1.18%	-0.53%	0.32%	1.07%
FF3	10,333	0.32%	1.09%	-0.40%	0.30%	1.01%
FFC4	10,333	0.12%	1.56%	-0.98%	-0.12%	1.06%
CAPM REIT	10,333	0.08%	0.47%	-0.11%	0.09%	0.30%
FF3 REIT	10,333	0.10%	0.47%	-0.09%	0.10%	0.31%
FFC4 REIT	10,333	0.06%	0.68%	-0.27%	0.10%	0.44%
Net Return Variables						
Net Returns	10,333	0.72%	1.65%	0.04%	0.86%	1.75%
CAPM	10,333	0.13%	1.18%	-0.64%	0.22%	0.96%
FF3	10,333	0.21%	1.09%	-0.51%	0.19%	0.90%
FFC4	10,333	0.01%	1.56%	-1.08%	-0.22%	0.96%
CAPM REIT	10,333	-0.02%	0.47%	-0.22%	-0.02%	0.19%
FF3 REIT	10,333	-0.01%	0.47%	-0.21%	-0.01%	0.20%
FFC4 REIT	10,333	-0.05%	0.68%	-0.38%	-0.01%	0.34%
Fund Flows (%)	10,333	-0.65%	6.71%	-4.31%	-1.22%	2.41%
Morningstar Ratings	6,772	2.76	0.97	2.00	2.92	3.50
Panel C: Institutional sample						
Variable	N	Mean	Std Dev	Q1	Median	Q3
Gross Return Variables						
Gross Returns	7,677	0.88%	1.55%	0.17%	0.93%	1.84%
CAPM	7,677	0.06%	1.18%	-0.64%	0.18%	0.88%
FF3	7,677	0.21%	1.17%	-0.56%	0.21%	0.98%
FFC4	7,677	0.02%	1.56%	-1.08%	-0.18%	0.98%
CAPM REIT	7,677	0.10%	0.50%	-0.11%	0.09%	0.29%
FF3 REIT	7,677	0.14%	0.52%	-0.08%	0.11%	0.31%
FFC4 REIT	7,677	0.11%	0.73%	-0.27%	0.09%	0.46%
Net Return Variables						
Net Returns	7,677	0.79%	1.55%	0.09%	0.86%	1.76%
CAPM	7,677	-0.02%	1.18%	-0.72%	0.10%	0.79%
FF3	7,677	0.13%	1.17%	-0.64%	0.13%	0.90%
FFC4	7,677	-0.07%	1.56%	-1.16%	-0.26%	0.90%
CAPM REIT	7,677	0.02%	0.50%	-0.19%	0.02%	0.20%
FF3 REIT	7,677	0.06%	0.52%	-0.17%	0.03%	0.23%
FFC4 REIT	7,677	0.03%	0.73%	-0.36%	0.01%	0.38%
Fund Flows (%)	7,677	-1.19%	6.32%	-4.43%	-1.43%	1.98%
Morningstar Ratings	3,642	3.10	0.98	2.25	3.00	3.83

Table 2: Correlation matrix

This table presents Pearson correlation coefficients for each of the performance metrics in the study. All performance metrics in this table are on a net return basis, as the gross return variables are similar. CAPM, FF3, and FFC4 refer to traditional factor models as defined in the text, and CAPM REIT, FF3 REIT, and FFC4 REIT refer to REIT-based factor models.

Panel A: Full Sample								
Performance Measure	Morningstar	Net Returns	CAPM	FF3	FFC4	CAPM REIT	FF3 REIT	FFC4 REIT
Morningstar	1.00	0.07	0.06	0.03	0.02	0.26	0.19	0.14
Net Returns	0.07	1.00	0.61	0.34	0.40	0.38	0.31	0.15
CAPM	0.06	0.61	1.00	0.72	0.43	0.33	0.23	0.25
FF3	0.03	0.34	0.72	1.00	0.70	0.26	0.20	0.21
FFC4	0.02	0.40	0.43	0.70	1.00	0.24	0.16	0.08
CAPM REIT	0.26	0.38	0.33	0.26	0.24	1.00	0.83	0.63
FF3 REIT	0.19	0.31	0.23	0.20	0.16	0.83	1.00	0.72
FFC4 REIT	0.14	0.15	0.25	0.21	0.08	0.63	0.72	1.00

Table 3: Flows to top and bottom-ranked funds

This table presents average fund flows to top and bottom-quartile funds ranked by each performance metric. All alphas and returns are on a net of fees, as gross returns and alphas were similar. CAPM, FF3, and FFC4 refer to traditional factor models as defined in the text, and CAPM REIT, FF3 REIT, and FFC4 REIT refer to REIT-based factor models.

Panel A: Combined Sample

Morningstar Sample (N = 10,414)				Full Sample (N = 18,010)			
Performance Measure	Top	Bottom	Difference	Performance Measure	Top	Bottom	Difference
RETURNS	-0.16	-1.86	1.70	CAPM REIT	0.03	-1.68	1.71
MORNINGSTAR	-0.06	-1.75	1.59	RETURNS	0.01	-1.69	1.70
CAPM REIT	-0.17	-1.63	1.47	CAPM	0.01	-1.69	1.70
CAPM	-0.19	-1.65	1.46	FF3 REIT	-0.21	-1.63	1.42
FF3 REIT	-0.39	-1.54	1.15	FF3	-0.22	-1.61	1.38
FF3	-0.38	-1.49	1.11	FFC4 REIT	-0.26	-1.47	1.21
FFC4	-0.58	-1.53	0.94	FFC4	-0.35	-1.55	1.20
FFC4 REIT	-0.54	-1.48	0.94				

Table 3 – Cont'd

<i>Panel B: Retail sample</i>							
Morningstar Sample (N = 6,772)				Full Sample (N = 10,333)			
Performance Measure	Top	Bottom	Difference	Performance Measure	Top	Bottom	Difference
RETURNS	-0.12	-1.66	1.54	RETURNS	0.20	-1.51	1.71
MORNINGSTAR	-0.15	-1.39	1.24	CAPM	0.05	-1.54	1.59
CAPM	-0.36	-1.44	1.08	CAPM REIT	0.06	-1.40	1.47
CAPM REIT	-0.34	-1.35	1.01	FF3	-0.15	-1.35	1.21
FF3	-0.45	-1.28	0.82	FF3 REIT	-0.17	-1.36	1.19
FF3 REIT	-0.48	-1.30	0.82	FFC4 REIT	-0.14	-1.14	1.00
FFC4	-0.55	-1.30	0.75	FFC4	-0.30	-1.30	0.99
FFC4 REIT	-0.52	-1.24	0.72				

<i>Panel C: Institutional sample</i>							
Morningstar Sample (N = 3,642)				Full Sample (N = 7,677)			
Performance Measure	Top	Bottom	Difference	Performance Measure	Top	Bottom	Difference
CAPM REIT	-0.02	-2.39	2.36	CAPM REIT	-0.06	-2.08	2.14
CAPM	-0.06	-2.26	2.20	CAPM	-0.10	-2.03	1.93
RETURNS	-0.22	-2.39	2.17	FF3 REIT	-0.29	-1.99	1.71
MORNINGSTAR	-0.12	-2.22	2.10	RETURNS	-0.27	-1.94	1.68
FF3	-0.36	-2.09	1.72	FFC4 REIT	-0.42	-1.99	1.57
FF3 REIT	-0.48	-2.14	1.67	FF3	-0.44	-1.91	1.47
FFC4 REIT	-0.50	-1.96	1.46	FFC4	-0.44	-1.91	1.47
FFC4	-0.69	-1.95	1.25				

Table 4: Horse race tests

This table presents a series of horse race tests for two performance metrics. We estimate panel regressions of fund flows on indicators of whether a fund is a top-ranked fund by each performance measure and controlling for month fixed effects. We report the difference in the coefficients of the first and the second indicators with the t-statistic in parentheses. Standard errors are clustered by fund and month. CAPM, FF3, and FFC4 refer to traditional factor models as defined in the text, and CAPM REIT, FF3 REIT, and FFC4 REIT refer to REIT-based factor models.

Panel A: Combined Sample

Model Comparison	Difference	T-Stat	Model Comparison	Difference	T-Stat
Net Returns vs. All			Morningstar vs. All		
Net Returns vs. Morningstar	-0.22	-0.70	Morningstar vs Net returns	0.22	0.70
Net Returns vs. CAPM	0.15	0.59	Morningstar vs CAPM	0.12	0.34
Net returns vs FF3	0.63	3.74	Morningstar vs FF3	0.32	0.95
Net returns vs FFC4	0.61	3.29	Morningstar vs FFC4	0.52	1.54
Net returns vs CAPM REIT	0.08	0.39	Morningstar vs CAPM REIT	-0.04	-0.12
Net returns vs FF3 REIT	0.47	2.23	Morningstar vs FF3 REIT	0.33	0.91
Net returns vs FFC4 REIT	0.70	4.19	Morningstar vs FFC4 REIT	0.46	1.33
CAPM vs. All			CAPM REIT vs. All		
CAPM vs Net returns	-0.15	-0.59	CAPM REIT vs Net returns	-0.08	-0.39
CAPM vs Morningstar	-0.12	-0.34	CAPM REIT vs Morningstar	0.04	0.12
CAPM vs FF3	0.69	2.52	CAPM REIT vs CAPM	0.05	0.16
CAPM vs FFC4	0.57	1.37	CAPM REIT vs FF3	0.52	2.09
CAPM vs CAPM REIT	-0.05	0.16	CAPM REIT vs FFC4	0.52	2.58
CAPM vs FF3 REIT	0.41	1.61	CAPM REIT vs FF3 REIT	0.68	2.15
CAPM vs FFC4 REIT	0.64	2.56	CAPM REIT vs FFC4 REIT	0.87	3.79

Table 4 – Cont'd

<i>Panel B: Retail sample</i>					
Model Comparison	Difference	T-Stat	Model Comparison	Difference	T-Stat
Net Returns vs. All			Morningstar vs. All		
Net Returns vs. Morningstar	0.07	0.18	Morningstar vs Net returns	-0.07	-0.18
Net Returns vs. CAPM	0.76	2.15	Morningstar vs CAPM	0.12	0.32
Net returns vs FF3	0.85	3.34	Morningstar vs FF3	0.51	1.41
Net returns vs FFC4	0.83	2.33	Morningstar vs FFC4	0.68	1.60
Net returns vs CAPM REIT	0.74	2.01	Morningstar vs CAPM REIT	0.20	0.47
Net returns vs FF3 REIT	0.78	2.25	Morningstar vs FF3 REIT	0.54	1.15
Net returns vs FFC4 REIT	0.97	3.29	Morningstar vs FFC4 REIT	0.59	1.45
CAPM vs. All			CAPM REIT vs. All		
CAPM vs Net returns	-0.76	-2.15	CAPM REIT vs Net returns	-0.74	-2.01
CAPM vs Morningstar	-0.12	-0.32	CAPM REIT vs Morningstar	-0.20	-0.47
CAPM vs FF3	0.51	1.64	CAPM REIT vs CAPM	0.14	0.40
CAPM vs FFC4	0.48	1.60	CAPM REIT vs FF3	0.48	0.93
CAPM vs CAPM REIT	-0.14	-0.40	CAPM REIT vs FFC4	0.50	1.99
CAPM vs FF3 REIT	0.42	1.12	CAPM REIT vs FF3 REIT	0.82	1.52
CAPM vs FFC4 REIT	0.64	1.98	CAPM REIT vs FFC4 REIT	0.95	2.42

Table 4 – Cont'd

<i>Panel C: Institutional Sample</i>					
Model Comparison	Difference	T-Stat	Model Comparison	Difference	T-Stat
Net Returns vs. All			Morningstar vs. All		
Net Returns vs. Morningstar	-0.16	-0.41	Morningstar vs Net returns	0.16	0.41
Net Returns vs. CAPM	0.85	0.98	Morningstar vs CAPM	0.16	0.76
Net returns vs FF3	1.10	1.76	Morningstar vs FF3	0.39	0.23
Net returns vs FFC4	1.22	1.73	Morningstar vs FFC4	0.75	0.71
Net returns vs CAPM REIT	-0.56	-1.86	Morningstar vs CAPM REIT	-0.52	-0.90
Net returns vs FF3 REIT	1.13	0.70	Morningstar vs FF3 REIT	0.64	0.29
Net returns vs FFC4 REIT	1.30	1.46	Morningstar vs FFC4 REIT	0.65	0.35
CAPM vs. All			CAPM REIT vs. All		
CAPM vs Net returns	-0.85	-0.98	CAPM REIT vs Net returns	0.56	1.86
CAPM vs Morningstar	-0.16	-0.76	CAPM REIT vs Morningstar	0.52	0.90
CAPM vs FF3	0.70	1.28	CAPM REIT vs CAPM	0.42	2.20
CAPM vs FFC4	0.68	1.24	CAPM REIT vs FF3	0.37	1.87
CAPM vs CAPM REIT	-0.42	-2.20	CAPM REIT vs FFC4	0.38	2.14
CAPM vs FF3 REIT	0.70	1.27	CAPM REIT vs FF3 REIT	0.55	1.94
CAPM vs FFC4 REIT	0.86	1.88	CAPM REIT vs FFC4 REIT	0.74	2.87

Table 5: Determinants of aggregate flows to active funds

This table presents time series regressions of monthly aggregate flows to active funds onto monthly aggregate fund performance measured as of the end of the previous month. Eight measures of fund performance are considered: net returns, Morningstar ratings, and alphas from the CAPM model, the three-factor Fama-French model, the four factor Fama-French-Carhart model, REIT CAPM model, the three-factor REIT Fama-French model, and the REIT four factor Fama-French-Carhart model. To facilitate the comparison of effects, we measure the dependent variable in percentage points, and all continuous explanatory variables are transformed into a standard score. The specifications include calendar-month fixed effects. T-statistics based on Newey-West standard errors with a six-lag correction are presented in parentheses. Panel A reports the results for the *Retail Sample*, with Net Returns as the focal variable. Panel B reports the results for the *Institutional Sample* with REIT CAPM as the focal variable.

<i>Panel A: Retail Sample - Net Returns vs All</i>								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Net Returns</i>	0.010** (2.28)	0.010** (2.24)	0.003 (0.51)	0.007* (1.69)	0.011** (2.10)	0.013*** (3.02)	0.012*** (3.16)	0.010** (2.34)
<i>Morningstar</i>		0.000 (0.03)						
<i>CAPM</i>			0.016 (1.47)					
<i>FF3</i>				0.006 (1.62)				
<i>FFC4</i>					-0.003 (-0.69)			
<i>REIT CAPM</i>						-0.006* (-1.96)		
<i>REIT FF3</i>							-0.008** (-2.27)	
<i>REIT FFC4</i>								-0.002 (-0.75)
<i>Calendar month FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Number of months</i>	325	325	325	325	325	325	325	325
<i>Adj R2</i>	0.037	0.034	0.010	0.047	0.036	0.046	0.055	0.036

Table 5 – Cont'd

<i>Panel B: Institutional Sample - CAPM REIT vs All</i>								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CAPM REIT	0.008** (2.35)	0.008** (2.38)	0.011** (2.34)	0.008** (2.39)	0.011*** (2.79)	0.012*** (3.35)	0.008** (2.42)	0.008*** (2.76)
Morningstar		-0.005 (-1.45)						
Net Returns			-0.005 (-0.95)					
CAPM				-0.003 (-0.67)				
FF3					-0.004 (-1.25)			
FFC4						-0.008*** (-2.69)		
REIT FF3							-0.002 (-0.61)	
REIT FFC4								-0.006** (-2.20)
Calendar month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of months	325	325	325	325	325	325	325	325
Adj R ²	0.066	0.076	0.070	0.065	0.068	0.086	0.065	0.079

Table 6: Horse race retail vs. institutional categorization robustness

This table presents a series of horse race tests for two performance metrics. We estimate panel regressions of fund flows on indicators of whether a fund is a top-ranked fund by each performance measure and controlling for month fixed effects. We report the difference in the coefficients of the first and the second indicators with the t-statistic in parentheses. Standard errors are clustered by fund and month. CAPM, FF3, and FFC4 refer to traditional factor models as defined in the text, and CAPM REIT, FF3 REIT, and FFC4 REIT refer to REIT-based factor models. We report the results using various thresholds for the Retail/Institutional classification. Funds are assigned to the retail category if the proportion of shares held by retail investors exceeds 40%, 50%, and 60% thresholds, respectively.

<i>Panel A: Retail Sample</i>						
<i>Cutoff</i>	<i>>40% (N=11,043)</i>		<i>>50% (N=9,555)</i>		<i>>60% (N=8,900)</i>	
Model Comparison	Difference	T-Stat	Difference	T-Stat	Difference	T-Stat
Net Returns vs. All						
Net returns vs CAPM REIT	0.79	2.29	0.74	2.01	0.78	2.26
Net Returns vs. Morningstar	0.03	0.08	0.07	0.18	-0.06	0.15
Net Returns vs. CAPM	0.72	2.09	0.76	2.15	0.72	2.11
Net returns vs FF3	0.91	4.33	0.85	3.34	0.77	3.22
Net returns vs FFC4	0.82	2.65	0.83	2.33	0.98	2.68
Net returns vs FF3 REIT	0.78	2.42	0.78	2.25	0.80	2.42
Net returns vs FFC4 REIT	1.08	3.55	0.97	3.29	1.10	3.75
<i>Panel B: Institutional Sample</i>						
<i>Cutoff</i>	<i>>40% (N=9,110)</i>		<i>>50% (N=7,677)</i>		<i>>60% (N=6,967)</i>	
Model Comparison	Difference	T-Stat	Difference	T-Stat	Difference	T-Stat
CAPM REIT vs. All						
CAPM REIT vs Net returns	0.60	1.89	0.56	1.86	0.69	2.17
CAPM REIT vs Morningstar	0.42	0.12	0.52	0.90	0.62	1.10
CAPM REIT vs CAPM	-0.01	0.16	0.42	2.20	0.52	2.08
CAPM REIT vs FF3	0.23	1.09	0.37	1.87	0.96	1.95
CAPM REIT vs FFC4	0.47	2.58	0.38	2.14	0.86	2.62
CAPM REIT vs FF3 REIT	0.51	2.15	0.55	1.94	0.57	1.91
CAPM REIT vs FFC4 REIT	0.55	3.79	0.74	2.87	0.74	2.08

Table 7: Aggregate flows retail vs. institutional categorization robustness

This table presents time series regressions of monthly aggregate flows to active funds onto monthly aggregate fund performance measured as of the end of the previous month. Eight measures of fund performance are considered: net returns, Morningstar ratings, and alphas from the CAPM model, the three-factor Fama-French model, the four factor Fama-French-Carhart model, REIT CAPM model, the three-factor REIT Fama-French model, and the REIT four factor Fama-French-Carhart model. To facilitate the comparison of effects, we measure the dependent variable in percentage points, and all continuous explanatory variables are transformed into a standard score. The specifications include calendar-month fixed effects. T-statistics based on Newey-West standard errors with a six-lag correction are presented in parentheses. Panel A reports the results for the *Retail Sample*, with Net Returns as the focal variable. Panel B reports the results for the *Institutional Sample* with REIT CAPM as the focal variable. Funds are assigned to the retail category if the proportion of shares held by retail investors exceeds 40%, 50%, and 60% thresholds, respectively.

<i>Panel A: Retail Sample - Net Returns vs Other Metrics</i>									
	>40%			>50%			>60%		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Net Returns	0.007** (1.98)	0.008** (1.99)	0.011*** (2.96)	0.010** (2.28)	0.010** (2.24)	0.013*** (3.02)	0.010** (2.43)	0.010** (2.37)	0.013*** (3.28)
Morningstar		-0.001 (-0.25)			0.000 (0.03)			-0.000 (-0.01)	
CAPM REIT			-0.007** (-2.02)			-0.006* (-1.96)			-0.006 (-1.64)
N	325	325	325	325	325	325	325	325	325
<i>Panel B: Institutional Sample - CAPM REIT vs Other Metrics</i>									
	>40%			>50%			>60%		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CAPM REIT	0.008** (2.44)	0.008** (2.54)	0.011** (2.36)	0.008** (2.35)	0.008** (2.38)	0.011** (2.34)	0.009*** (2.72)	0.008*** (2.60)	0.012*** (2.77)
Morningstar		-0.005 (-1.40)			-0.005 (-1.45)			-0.006** (-2.15)	
Net Returns			-0.005 (-0.86)			-0.005 (-0.95)			-0.005 (-0.88)
N	325	325	325	325	325	325	325	325	325

Table 8: Horse race alpha calculation robustness

Panel A (Panel B) presents a series of horse race tests between Net Returns (CAPM REIT) and other performance metrics at a time. We estimate panel regressions of fund flows on indicators of whether a fund is a top-ranked fund by each performance measure and controlling for month fixed effects. In Panel A (Panel B), we report the t-stat of the difference in the coefficients between Net Returns (CAPM REIT). Standard errors are clustered by fund and month. CAPM, FF3, and FFC4 refer to traditional factor models as defined in the text and CAPM REIT, FF3 REIT, and FFC4 REIT refer to REIT-based factor models. We present the results of our test employing various computation methodologies including alpha computed using 12, 24, and 36 months rolling window, and alpha moving averages computed using 6, 12, and 18 months intervals.

<i>Panel A: Retail Sample</i>		Horse race of T-stat of Net Returns vs. Other Metrics						
Alpha Window	Average Months	CAPM REIT	Morningstar	CAPM	FF3	CAT4	FF3 REIT	CAT4 REIT
12	1	0.87	0.20	0.66	0.26	2.38	0.67	0.53
24	1	0.61	1.97	0.11	0.42	0.05	0.19	0.33
36	1	0.59	2.32	0.23	0.27	0.54	0.22	0.19
12	6	2.21	0.77	2.82	0.91	0.19	0.58	2.83
24	6	2.39	0.44	2.27	0.75	0.00	0.47	1.95
36	6	1.04	0.36	2.03	0.13	1.11	0.19	1.13
12	12	2.01	0.18	2.15	3.34	2.33	2.25	3.29
24	12	2.34	0.11	1.17	2.23	1.88	2.49	2.61
36	12	2.12	0.21	1.66	1.86	1.17	1.91	1.86
12	18	2.87	0.42	2.47	2.83	2.53	2.50	2.81
24	18	2.83	0.79	1.72	2.70	2.12	2.31	1.80
36	18	1.93	0.39	1.00	1.96	1.14	1.82	1.19

<i>Panel B: Institutional Sample</i>		Horse race of T-stat of CAPM REIT vs. Other Metrics						
Alpha Window	Average Months	Net returns	Morningstar	CAPM	FF3	CAT4	FF3 REIT	CAT4 REIT
12	1	0.31	2.48	0.28	0.35	0.38	0.13	0.59
24	1	0.44	2.71	0.19	0.17	0.69	1.30	0.68
36	1	3.36	0.72	0.61	0.08	1.09	0.20	2.59
12	6	1.79	0.31	2.25	0.91	0.23	1.86	2.06
24	6	1.45	0.83	2.08	0.93	0.09	1.64	2.02
36	6	0.41	0.82	0.08	0.30	0.42	0.50	0.19
12	12	1.86	0.90	2.20	1.87	2.14	1.94	2.87
24	12	2.05	0.21	2.23	1.80	1.87	1.88	2.82
36	12	1.68	0.56	1.82	1.67	2.16	1.64	1.95
12	18	2.11	1.12	1.98	1.81	2.35	2.19	2.69
24	18	1.14	0.82	1.84	1.31	2.64	1.74	2.63
36	18	0.05	0.27	0.39	0.74	1.16	0.27	1.00

Appendix Table 9: Literature summary of REMF performance and benchmark models

Paper	Data Source	Sample Period	Performance	Benchmark model ¹⁰
Chou and Hardin (2014)	CRSP	1994-2006	Zero	Fama French Carhart model with REIT based Factors and SIC code factors
O'Neal and Page (2000)	Morningstar	1996-1998	Zero	Custom multifactor market model (REIT, Russel 2000, S&P500, World)
Kallberg, Liu, and Trzcinka (2000)	Morningstar	1987-1998	Positive	Custom multifactor model (size, value, bond) with REIT index as market factor
Lantushenko and Nelling (2019)	CRSP	1995-2015	Zero	Fama French Carhart Model with REIT based factors and traditional Factors
Lin and Yung (2004)	WSJ and Yahoo	1993-2001	Negative	Fama French Carhart Model with REIT index as market factor
MacGregor, Schulz, and Zhao (2020)	CRSP	1992-2016	Zero	Fama French Carhart Model + Liquidity factor using traditional factors
Kaushik and Pennathur (2012)	CRSP	1990-2008	Positive	Fama French Carhart Model with REIT as market factor
Hartzell, Mulhlofer, and Titman (2010)	CRSP	1994-2005	Zero	Fama French Carhart Model with REIT based factors + property type factors + non-REIT factors
Derwall, Huij, Brounen, and Marquering (2009)	CRSP	1980-2008	Zero	Fama French Carhart Model with REIT based momentum and traditional factors for the others
Chiang, Kozhevnikov, Lee, and Wisen (2008)	Morningstar	1982-2003	Zero	Fama French 3-Factor Model with traditional factors
Gallo, Lockwood, and Rutherford (2000)	Morningstar	1991-1997	Positive	CAPM with REIT index as market factor
Rodriguez (2007)	CRSP/Morningstar	1999-2004	Zero	CAPM with REIT index as market factor

¹⁰ Several papers use more than one benchmark model. This column represents the factor model used with the most factors. For example, Lantushenko and Nelling (2019) also use traditional CAPM models and CAPM models with a REIT index as a marked

